Q2:- Define Schwarz's theorem and Proof

$|X-Z| \leq |X-Y| + |Y-Z|$

Sol:- Schwarz's triangle Inequality states that for any three points X, Y, Z in a metric Space, the following Inequality holds.

$|X-Z| \leq |X-Y| + |Y-Z|$

Proof: we want to Show that using the triangle Inequality for distances, we have

$|X-Y| \geq |X-Z| - |Y-Z|$ ——— (1)

Similarly, Apply Again triangle Inequality

$|Y-Z| \geq |X-Z| - |X-Y|$ ——— (2)

Now, add these two Inequalities

$|X-Y| + |Y-Z| \geq 2|X-Z| - |X-Y| - |Y-Z|$

plify $2|X-Z| \leq |X-Y| + |Y-Z| + |X-Y| + |Y-Z|$

nbine $2|X-Z| \leq 2|X-Y| + 2|Y-Z|$

Divided b/s by 2

$|X-Z| \leq |X-Y| + |Y-Z|$

Thus, we have Proved Schwarz's Triangle Inequality.

Starting from Maclaurin Series $\log(1-x) = \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}$, $|x|<1$. Show by Abel's Theorem

$$\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)}.$$

Sol:- Abel's theorem State that if a power Series $\sum_{n=0}^{\infty} a_n x^n$ converges at $x=r$ and Diverg $x=R$.

In this case, we have the power series $\sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}$ which converges for $|x|<1$.

Start find radius of Convergence

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left| \frac{\frac{x^{n+2}}{n+2}}{\frac{x^{n+1}}{n+1}} \right| = \lim_{n\to\infty} \left|\frac{x}{n+2}\right| = 0$$

Since the limit less than 1 radius of convergen $R=1$

Using Abel's theorem

$$\sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = \lim_{x\to 1^-} \frac{x^{n+1}}{n+1} = \lim_{n\to 1^-} \left(\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} \right) - \lim_{n\to 1} \frac{x}{1+1}$$

$$\sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = \log(1-x) - \frac{1}{2}$$

$$\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} = \log(1-x) - \frac{1}{2}$$

the series $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)}$ is equal to $\log(1-1) - \frac{1}{2}$.

Find 1st Four terms by maclaurin Series

$$\frac{e^{-x^2}}{1+x^2}$$

Given $\frac{e^{-x^2}}{1+x^2}$

Maclaurin Series for e^{-x^2}

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$$

Put

$$\frac{e^{-x^2}}{1+x^2} = \frac{1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots}{1+x^2}$$

Divide & Simplify

$$\frac{e^{-x^2}}{1+x^2} = \left(1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots\right) \cdot \left(\frac{1}{1+x^2}\right).$$

Date: / /20 Day:

Prove that $\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right) \cdot = \int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right) dx = 0$.

We use symmetry Property of Sine & cosine fun

Integral sine func

$$\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right) dx$$

Sine fn is an odd fun ($\sin(-\theta) = -\sin\theta$), and the limits of Integration is zero.

Similarly for Cosine:

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right) dx$$

Cosine function is even function ($\cos(-\theta) = \cos\theta$)

due to symmetry the limits around zero

Therefore both Integrals are equal to zero

$$\int_{-L}^{L} \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

$$\int_{-L}^{L} \cos\left(\frac{n\pi x}{L}\right) dx = 0 .$$

What is the Schwarz inequality. Further Prove $|x-z| \leq |x-y| + |y-z|$ for any x, y and z in R^n.

Sol:- The Schwarz Inequality also known as the triangle Inequality, states that for any vectors x and y in R^n, the following holds:

$$|x \cdot y| \leq \|x\| \cdot \|y\|$$

To prove $|x-z| \leq |x-y| + |y-z|$ for any x, y, z in R^n, Consider the triangle Inequality for vectors. Let $a = x-y$ & $b = y-z$. Then

$$|x-z| = |(x-y)+(y-z)| \leq |x-y|+|y-z|$$

Required Solution.

Therefore $|x-z| \leq |x-y|+|y-z|$ hold for any x, y, z in R^n.

Show that limit of a Continous function is discontinous on R where

$$f_n(x) = \frac{x^{2n}}{1+x^{2n}}$$

Let's consider the pointwise limit of $f_n(x)$ as n approaches infinity

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{x^{2n}}{1+x^{2n}}$$

we can divid b/s x^{2n}

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{1}{\frac{1}{x^{2n}}+1}$$

Now as $n\to\infty$, $\frac{1}{x^{2n}}$ approaches zero if $|x|>1$ and it approaches infinity if $|x|<1$.

Case I

$|x|>1$ as $n\to\infty$, $\frac{1}{x^{2n}}$ approach zero

$$\lim_{n\to\infty} f_n(x) = \frac{1}{0+1} = 1$$

Case II

$|x|<1$ as $n\to\infty$, $\frac{1}{x^{2n}}$ approach Infinity.

$$\lim_{n\to\infty} f_n(x) = \frac{1}{\infty+1} = 0$$

Q. Find the radius of Convergence of the Power series $\sum_{n=1}^{\infty} 2^n n^p (x+1)^n$.

Sol: The ratio test States if

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L \text{ , then}$$

$a_n = 2^n n^p (x+1)^n$

Apply Ratio-Test

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{2^{n+1}(n+1)^p (x+1)^{n+1}}{2^n n^p (x+1)^n}\right|$$

Simplifying the expression

$$\lim_{n\to\infty} \left|\frac{2(n+1)(x+1)}{n}\right|.$$

Take absolute

$$\lim_{n\to\infty} \frac{2(n+1)|x+1|}{n}$$

Now, Find $\lim n\to\infty$

$$L = \frac{2|x+1|(\infty+1)}{\infty} = 2|x+1| = L$$

Radius of Convergence $R = \frac{1}{L}$

$\frac{1}{2|x+1|}$ is the radius of Convergence.

Find the set S on which $\{f_{(n)}(x)\}$ converges to Pointwise and find the limit function where $f_n(x) = nx^n(1-x^2)$.

The sequence $\{f_n(x)\}$ is defined as

$f_n(x) = nx^n(1-x^2)$.

let's limit of $f_n(x)$

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} nx^n(1-x^2)$$

For $x=0$,

$$\lim_{n\to\infty} f_n(0) = \lim_{n\to\infty} n\cdot 0^n(1-0^2) = 0$$

For $x=1$,

$$\lim_{n\to\infty} f_n(1) = \lim_{n\to\infty} n\cdot 1^n(1-1^2) = 0$$

For $x=-1$,

$$\lim_{n\to\infty} f_n(-1) = \lim_{n\to\infty} n\cdot(-1)^n(1-(-1)^2)$$

Here limit Doesn't exist bcz $(-1)^n$ oscilate b/w $-1, 1$

For x such that $-1<x<0$ or $0<x<1$

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} nx^n(1-x^2) = \infty$$

Therefore, the sequence $\{f_n(x)\}$ converges Pointwise

$$f(x) = \begin{cases} 0 & \text{for } x = 0, 1 \\ \text{Undefined for } x = -1 \\ \infty & \text{for } -1<x<0 \text{ or } 0<x<1 \end{cases}$$

Hence, the set S on which $\{f_n(x)\}$ converges Pointwis is $S = \{0, 1\}$. Answer

If $\{f_n\}$ is uniformly Convergent Sequence of Continous function On $[0,1]$, then Prove that $\{f_n\}$ is equicontinous.

Since $\{f_n\}$ is uniformly Convergent on $[0,1]$, the limit function f it Converges to is also Continous

Now Consider any function f_n is sequence.

for a fixed $\varepsilon > 0$, there exist as N such that

$|f_n(x) - f(x)| < \varepsilon/3$ for all $n \geq N$ $x \in [0,1]$.

Now,

Due to the continuity of f, there exist a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon/3$ whenever $|x-y| < \delta$.

for $n \geq N$, using the triangle Inequality

$$|f_n(x) - f_n(y)| \leq |f_n(x) - f(x)| + |f(x) - f(y)| +$$

$$+ |f(y) - f_n(y)| < \varepsilon$$

Thus, for any $n \geq N$, $|f_n(x) - f_n(y)| < \varepsilon$ whenever $|x-y| < \delta$,

Proving equicontinuity.

Given the Series $\sum_{n=1}^{\infty} f_n$ for which

$S_n(x) = \frac{1}{2n^2} \log(1 + n^4x^2)$; $0 \le x \le 1$

Show that the series $\sum_{n=1}^{\infty} f'_n$ does not converges uniformly but the given Series Can be differentiated term by term.

Let's find the derivative of $S_n(x)$ using chain rule

Given: $S_n(x) = \frac{1}{2n^2} \log(1 + n^4x^2)$

Now Derivative

$$S'_n(x) = \frac{d}{dx}\left(\frac{1}{2n^2} \log(1 + n^4x^2)\right)$$

Apply chain rule

$$S'_n(x) = \frac{1}{2n^2} \cdot \frac{1}{1 + n^4x^2} \cdot 2n^4x$$

$$S'_n(x) = \frac{n^2x}{1 + n^4x^2}$$

Partial sum of the derivative series

$$S(x) = \sum_{n=1}^{\infty} S'_n(x)$$

this serie doesn't converge uniformly

Let choose $x = 1$ the Partial Sums becomes

$$S(1) = \sum_{n=1}^{\infty} \frac{n^2}{1 + n^4}$$

Difference b/w two consecutive Partial Sums.

$$|S_{n+1}(1) - S_n(1)| = \left| \frac{(n+1)^2}{1 + (n+1)^4} - \frac{n^2}{1 + n^4} \right|$$

$$|S_{n+1}(1) - S_n(1)| = \frac{2n^5 + 2n^4 + 2n^3 + 2n^2 + 2n + 1}{(1 + n^4)(1 + (n+1)^4)}$$

Prove $\sum_{n=1}^{\infty} f'$ doesn't converge

Show $a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx$ for $n=1,2,3,4$.

Show limit of a Continous function is Discontinous on R where $f_n(x) = \frac{x^{2n}}{1+x^{2n}}$

Given:

$$f_n(x) = \frac{x^{2n}}{1+x^{2n}}$$

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{x^{2n}}{1+x^{2n}}$$

x^{2n} divide numerator and Denominator

$$\lim_{n\to\infty} f_n(x) = \lim \frac{1}{\frac{1}{x^{2n}}+1}$$

Now,

As n approach ∞, $\frac{1}{x^{2n}}$ aproach 0 if $|x|>1$, and it approaches Infinity if $|x|<1$.

Case I $|x|>1$ $n\to\infty$ the denominator approach 1.

$$\lim_{n\to\infty} f_n(x) = \frac{1}{1+1} = \frac{1}{2}$$

Case II. $|x|<1$ $n\to\infty$ the denominator approach ∞

$$\lim_{n\to\infty} f_n(x) = \frac{1}{\infty} = 0$$

the limit of $f_n(x)$ depend on value of x. if $|x|>1$, the limit is $\frac{1}{2}$ and if $|x|<1$ the limit is 0.

we Conclude that the limit of function is Discontinous on R.

Define the Identity by Maclaurin Series of $2\sin x\cos x = \sin 2x$.

Sol:- $2\sin x\cos x = \sin 2x$

We'll use the double angle Identity for Sin:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

&

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

Now

$$2\sin(x)\cos(x) = 2\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right)\left(1 - \frac{x^2}{2} + \frac{x^4}{4} - \frac{x^6}{6} + \cdots\right)$$

Now, expand the product keeping terms up to x^3.

$$2\sin(x)\cos(x) = 2x - \frac{2x^3}{3!} + \cdots$$

Now, Compare this with Maclaurin series of $\sin 2x$

$$\sin(2x) = 2x - \frac{(2x)^3}{3!} + \cdots$$

Both Series match up to cubic term and beyond - Confirming that $2\sin(x)\cos(x) = \sin(2x)$ based on the maclaurin Series -